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Bridging the Observability Gap with AI-Driven Insights

Cisco CX Hackathon 2026 · Team CS-IC-62

Demo

Watch the demo video

Problem Statement

AppDynamics environments continuously generate rich observability telemetry — health metrics, maturity scores, agent states, configuration gaps — but converting that raw data into *actionable customer guidance* is still a largely manual process:

1. **Time-to-insight is too slow.** Running an assessment produces multi-tab Excel and PowerPoint reports that engineers must manually open, interpret, and cross-reference before any recommendation reaches the customer.
2. **Expertise barrier is high.** Extracting meaningful conclusions requires deep familiarity with both the AppDynamics API surface and the maturity scoring model — knowledge that doesn't scale across the entire CS org.
3. **Reports go stale and get lost.** Assessment outputs are rarely archived consistently, making trend analysis and cross-customer pattern recognition nearly impossible.
4. **Follow-up questions restart the cycle.** Every clarifying question (“Which apps are blocking Gold tier?”) means another manual report review — or another tool run.

The result: observability data that *exists* but doesn't *flow*. The gap between raw telemetry and business-level guidance stays wide — and expensive.

Solution Overview

This project introduces an **AI-driven health check pipeline** that automates the full assessment lifecycle — from metric collection to report delivery to conversational Q&A — using the **Model Context Protocol (MCP)**, Cisco's **CIRCUIT LLM**, and a persistent **knowledge base** built from historical reports.

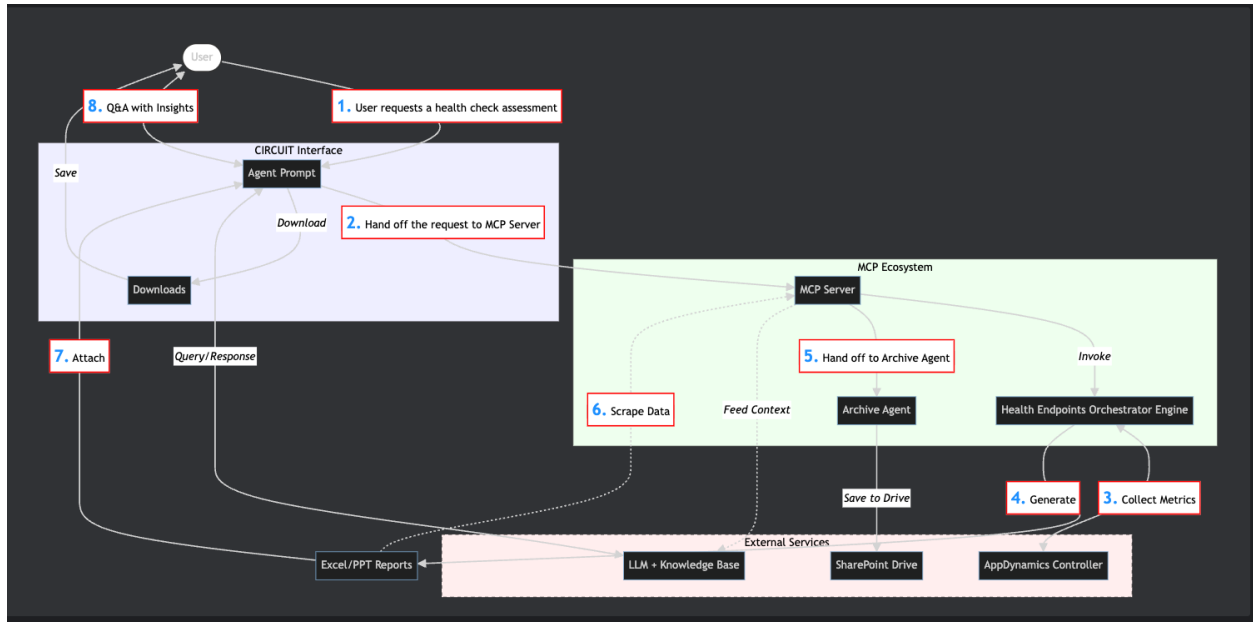


Figure 1: Advanced Health Check & LLM Insight Flow

Architecture

How the LLM powers each stage

Step	What happens	LLM's role
1 — Request	User asks for a health check in plain English via the CIRCUIT Agent Prompt	Parses intent, identifies target environment and job parameters
2 — Orchestration	CIRCUIT hands off a structured tool call to the MCP Server	Selects the correct MCP tool and arguments autonomously
3 — Metric collection	Health Endpoints Orchestrator Engine queries the AppDynamics Controller APIs	— (deterministic engine)
4 — Report generation	Engine produces Excel and PowerPoint reports	— (deterministic engine)
5 — Archiving	Archive Agent saves reports to SharePoint Drive for long-term retention	LLM can trigger archiving as part of the same conversation turn
6 — Knowledge ingestion	Report data is scraped and fed into the LLM's context / knowledge base	Builds cumulative understanding across runs and customers
7 — Delivery	Reports are attached to the CIRCUIT prompt and made available for download	LLM narrates key findings and highlights critical gaps
8 — Conversational Q&A	User asks follow-up questions; LLM answers from loaded context	Provides analyst-quality responses without re-running any tools

Why LLM is the right approach

- **Tool-use reasoning** — CIRCUIT receives the full MCP tool schema at session start and autonomously decides *which* tool to call, *when*, and with *what arguments*.
 - **Structured → natural translation** — Dense multi-sheet Excel data (maturity scores, agent matrices, threshold breaches) is converted into concise, human-readable summaries and prioritised recommendations.
 - **Persistent knowledge base** — Each run feeds the LLM's context, enabling cross-run and cross-customer pattern recognition that no single-session tool can provide.
 - **Conversational memory** — The full report is injected into context once. Follow-up questions are answered from memory — no re-running, no re-downloading.
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Setup & Run Steps

Prerequisites

- Python 3.12+
- Access to Cisco's **CIRCUIT LLM API** (app key, client ID, client secret)
- An AppDynamics controller + a valid CAT job file in `input/jobs/`
- SharePoint credentials (for Archive Agent, optional)

1 — Clone and install dependencies

```
git clone https://www.github.cisco.com/CSHackathon26/CS-IC-62-Bridging-the-Observability-Gap
cd CS-IC-62-Bridging-the-Observability-Gap-with-AI-Driven-Insights
```

```
pip install pipenv
pipenv install
pipenv shell
```

2 — Configure credentials

Set the following environment variables (or edit `properties.py` directly):

```
# Cisco CIRCUIT LLM
export CIRCUIT_LLM_API_APP_KEY=<your-app-key>
export CIRCUIT_LLM_API_CLIENT_ID=<your-client-id>
export CIRCUIT_LLM_API_CLIENT_SECRET=<your-client-secret>
export CIRCUIT_LLM_API_MODEL_NAME="gpt-5-nano"    # or gpt-4o-mini

# OAuth (defaults to Cisco's IdP - change only if using a different tenant)
export OAUTH_ENDPOINT="https://id.cisco.com/oauth2/default/v1/token"

CIRCUIT_LLM_API_ENDPOINT and CIRCUIT_LLM_API_VERSION are pre-configured in
properties.py.
```

3 — Add your AppDynamics job file

Create or edit `input/jobs/DefaultJob.json`:

```
{
  "controllerURL": "https://<your-controller>.saas.appdynamics.com",
  "username": "<user>@<account>",
  "password": "<password>",
  "account": "<account>",
  "applicationFilter": {
    "apm": { "isDefault": true }
  }
}
```

4 — Launch the AI assistant

```
python mcp_llm_client.py
```

The client will: 1. Spawn the MCP Server subprocess 2. Authenticate with Cisco’s OAuth IdP 3. Connect to CIRCUIT LLM with the full tool schema 4. Drop you into an interactive chat session

5 — Run an assessment and explore insights

User: Run the health check for DefaultJob and summarise the APM maturity results
LLM requested tool execution...

```
> Running run_assessment {'job_file': 'DefaultJob'}
```

```
> Saved report: DefaultJob-MaturityAssessment-apm.xlsx
```

CIRCUIT: Assessment complete. Overall APM maturity score is 68% (Silver).

3 applications are blocked from Gold: missing Business Transaction configuration,
Call Graph enabled on > 20% of nodes, and no custom dashboard detected.

User: Which apps are furthest from Gold?

CIRCUIT: Based on the loaded report, the top 3 apps with the most Gold-tier gaps are...

User: Archive this and send me the download link

LLM requested tool execution...

```
> Running archive_report {'job_file': 'DefaultJob', 'destination': 'sharepoint'}
```

CIRCUIT: Report archived to SharePoint. Download link: https://...

User: quit

Tip: The LLM will not re-run the assessment for follow-up questions about the same job — it answers from the already-loaded context, keeping sessions fast.

Key Outcomes

Outcome	Detail
10× faster time-to-insight	Assessment results that previously required 30–60 min of manual Excel review are summarised by the LLM in seconds.

Outcome	Detail
Zero tool expertise required	Any CS engineer or SA can request a health check and receive consultant-grade guidance through natural language.
Automated archiving	Every run is saved to SharePoint via the Archive Agent — creating a searchable, time-stamped record with no manual effort.
Cumulative knowledge	Report data is fed back into the LLM knowledge base, enabling cross-run trend analysis and cross-customer pattern recognition over time.
Conversational drill-down	Follow-up questions are answered from in-context memory — no re-running tools, no re-downloading reports.
Cisco-native & secure	Runs entirely over CIRCUIT (Cisco's internal AI gateway) — all data stays on-network, satisfying enterprise data-residency requirements.